

Patron User Guide

1. Starting and logging in / registering

1. Launch the application so that the **Patron Login** window appears.
2. To **register a new patron**:
 - Enter a username in the first text field and a password in the second.
 - Click **Register**.
 - If either field is empty, a message tells you to enter both username and password.
 - If a patron with the same username/password already exists, a message tells you to use Login instead.
 - On success, a new Patron (with a max checkout count of 3 and an initial fine amount) is created and added to the library's list.
3. To **log in as an existing patron**:
 - Enter your username and password and click **Login**.
 - If either field is empty, you are prompted to fill both.
 - If no matching patron is found, you are asked to complete registration first.
 - On success, the **Patron Menu** window opens and the login window closes.
4. To **switch to the librarian side**, click **Switch to Librarian Login**, which opens the librarian login window and closes the patron login.

2. Patron Menu – searching and viewing books

The Patron Menu shows: a list of book titles, a search text field, and buttons for **Check In**, **Check Out**, **Pay Fines**, **Search**, and **Back**.

- The list initially shows the titles of all books on all shelves in the library.
- To **search by title**:
 - Type part or all of a book title into the search field.
 - Click **Search**.
 - The list is cleared and repopulated with titles whose (lower-cased) titles contain your query.
 - If the query is empty, the full list is restored.

3. Patron Menu – checking books in and out

Checking in a book (returning):

1. Select a title from the list that you currently have checked out.
2. Click **Check In**.

- If nothing is selected, you are prompted to select a book.
- The program searches your personal list of checked-out books to find the title.
- If found, the book is marked available and removed from your list, and a confirmation message is shown; the list is refreshed.
- If you do not have that book, you are notified that you do not appear to have it checked out.

Checking out a book (borrowing):

1. Select a title from the list.
2. Click **Check Out**.
 - If nothing is selected, a message tells you to select a book first.
 - The system searches all shelves to find a **Book** with that title.
 - If the book is already checked out, you are informed that it is unavailable.
 - If it is available and you are below your maximum checkout count, the book is checked out to you and you see a success message; the list is refreshed.
 - If you already have the maximum number of books (3), a message asks you to check in some books first.

4. Paying fines

From Patron Menu, click **Pay Fines** to open the **Patron Payment** window.

- The window shows your current fine amount as a dollar value label.
- To **pay part or all of your fines**:
 - Enter a payment amount in the formatted text field.
 - Click **Pay**.
 - The program converts the text to a number and calls the patron's **pay** method, then updates the fine label and clears the input field.
 - If the input cannot be parsed as a number, you are prompted to enter a valid number.

To return to the Patron Menu, click **Back**, which recreates PatronMenu with the same patron and library and closes the payment window.

5. Logging out / switching to login

From Patron Menu, click **Back** to return to the Patron Login window and close the menu.

Librarian User Guide

1. Logging in / creating librarian accounts

Open the **Librarian Login** window (either at startup or via “Switch to Librarian Login” from the Patron Login).

- Enter username and password and click **Login**:
 - The system loops through the library’s list of librarians and calls `authenticate` on each.
 - On a successful match, the **Librarian Dashboard** opens and the login window closes.
 - If no librarian matches, a dialog shows “Invalid login”.
- To **create a new librarian**:
 - Click **Create Librarian**.
 - A dialog appears with fields for new username and password.
 - After confirming, if either field is empty, you are told that fields cannot be empty.
 - Otherwise, a new `Librarian` is created, added to the library, and a success message is shown.
- To **switch to the Patron Login**, click **Switch to Patron Login**, which opens the patron login window and closes the librarian login.

2. Librarian Dashboard – main navigation

After login, the **Librarian Dashboard** provides buttons for all librarian operations:

- **View Catalog** – browse the full catalog.
- **View Shelves** – browse shelves and their books.
- **Add Book** – create a new book record and place it on a shelf.
- **Add Shelf** – create a new shelf for organizing books.
- **Check Out Books** – process a checkout from a cart for the current librarian user.
- **Check In Books** – process check-ins.
- **Fine Management** – apply fines to patrons.
- **Logout** – close the dashboard and return to librarian login.

Each button opens another window and closes the dashboard.

3. Managing catalog and shelves

Viewing the catalog or a shelf:

- **View Catalog** opens **BookListPage** with `shelf == null`, showing the full catalog from `library.getCatalog()`.
- **View Shelves** opens **ViewShelvesPage**; from there, selecting a shelf can open **BookListPage** filtered to that shelf’s books.

In **BookListPage**:

- Each book row shows title, author, genre, and availability flag.
- If the current user is a Librarian, each row includes an **Add to Cart** button that adds that book to the user's cart.
- The **Back** button returns to the Dashboard (or ViewShelvesPage when viewing a single shelf).

Adding a new shelf (AddShelfPage):

- Open via **Add Shelf** from the dashboard.
- Fields: Shelf Name, Genre, Max Rows, Max Columns.
- All fields are required; if any are empty you see an error message.
- Rows and columns must be positive integers; invalid or non-numeric entries trigger validation messages.
- On success, a new **Shelf** is created and added to the library, you see "Shelf added successfully!", and the Dashboard reopens.

Adding a new book (AddBookPage):

- Open via **Add Book**.
- Fields: Title, Author, Genre, Year, plus a dropdown of existing Shelves populated from `library.getShelves()`.
- All text fields are required; missing values cause an error.
- Year must be a valid integer; otherwise you get a "Year must be a valid number" message.
- On success:
 - A new **Book** is created and added to the library catalog.
 - If a shelf is selected and not full, the book is also added to that shelf; if the shelf is full, you see an error and remain on the page.
 - A confirmation appears and you are returned to the dashboard.

4. Check-out, check-in, and fines (librarian side)

CheckoutPage (librarian cart checkout):

- Opened via **Check Out Books**.
- Displays a list of books currently in the librarian user's cart, each with title, author, genre, and a **Remove** button.
- Removing a book from the cart refreshes the page and shows a confirmation dialog.
- Clicking **Confirm Checkout** iterates through the cart, calls `checkOut` on each book for the current user, clears the cart, shows "Checkout complete!", and returns to the Dashboard.
- **Back** returns directly to the Dashboard without changing the cart.

CheckInPage:

- Opened via **Check In Books**; although the full implementation is not shown here, the pattern mirrors CheckoutPage with actions for returning books for the librarian user.

FineManagementPage:

- Opened via **Fine Management**.
- Shows a list of all patrons; you select a patron to manage.
- At the bottom, enter a numeric **Fine Amount** and click **Apply Fine**.
 - If no patron is selected, you are prompted to select one.
 - If the amount is non-numeric or not positive, you see a validation message.
 - On success, the specified amount is added as a fine to the selected patron and a confirmation dialog displays the amount and patron name.
- **Back** returns to the Dashboard.

Logging out:

- From the Dashboard, click **Logout** to close it and re-open the Librarian Login window.

Patron vs. Librarian GUI

Aspect	Patron GUI	Librarian GUI
Entry screen	PatronLogin – supports register and login for patrons. PatronLogin-8.java	LibrarianLogin – supports login and creation of librarian accounts. LibrarianLogin-26.java
Main menu	PatronMenu – list of book titles, search, check-in/out, pay fines. PatronMenu-10.java	LibrarianDashboard – navigation to catalog, shelves, add book/shelf, checkout, checkin, fines. LibrarianDashboard-25.java
Search capability	Title text search filtering list of all books. PatronMenu-10.java	Catalog browsing via View Catalog and View Shelves; no free-text search in dashboard itself. LibrarianDashboard-25.java+1
Checkout behavior	Patron checks books out directly from global list of titles. PatronMenu-10.java	Librarian uses a cart and a dedicated CheckoutPage to confirm checkout. CheckoutPage-22.java+1
Fine handling	PatronPayment lets patron pay down their own fines. PatronPayment-12.java	FineManagementPage lets librarian apply new fines to any patron. FineManagementPage-23.java
Account management from GUI	Patron can only create their own patron via registration. PatronLogin-8.java	Librarian can create new librarian accounts via dialog. LibrarianLogin-26.java
Role switching	Button to switch to Librarian Login. PatronLogin-8.java	Button to switch to Patron Login. LibrarianLogin-26.java